

Landon Holligan

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EDUCATION

University of Colorado Boulder

Aug 2023 – May 2027

Concurrent BS/MS in Aerospace Engineering

GPA: 3.75

- Dean's List (5 Semesters: Fall 2023 – Spring 2026)

MS Focus Area: Fluids, Structures, and Materials

EXPERIENCE

CU Sounding Rocket Lab Avionics Structures Lead – Mechanical/Aerospace Engineering

May 2025 - Present

- Leading avionics bay structural design for our space-shot rocket, using FEMAP quasi-static/random vibration analysis to specify structural bolts, shock isolators, and PCB mounting locations, validating survivability through launch, deployment, and landing.
- Engineered, manufactured, and assembled a precision 2-axis gimbal ground station, achieving 0.2° pointing accuracy through tolerance stack-up analysis.
- Spearheaded design of a one-meter radio dish and antenna fixture system. Iterated a 3D-printed truss-ribbed structure to meet a 2 mm surface roughness requirement, with PETG selected for desert thermal/solar exposure and a modular fixture enabling patch/helix antenna attachment to the outer dish.

SWARM-EX Cubesat Structures Lead – Mechanical/Aerospace Engineering

June 2026 - Present

- Developing a system-level structural FEA model in ANSYS, modeling CubeSat rail constraints and pulling hinge PSD response to guide a solar array breakout model.
- Matured dual-deploy hinge mechanism design, specifying torsion spring material/constraints and running thermal and structural analysis to help eliminate binding risks.
- Redesigned the solar array tie-down system to resolve a year-long deployment failure, directing manufacturing and assembly of flight-simulated hardware for an upcoming test campaign.

Tendeg Structural Analysis Internship – Mechanical/Aerospace Engineering

June 2026 - August 2026

- Built and validated a mass simulator finite element model for the company's gimbal assembly from scratch in FEMAP, performing bolted-joint analysis under sine sweep/random vibration loads leading to a shear pin redesign to push positive margins.
- Correlated FEM random vibration response to test data within 5%, isolating a failure source the team had misdiagnosed and redirected the qualification unit's troubleshooting efforts.
- Authored three customer-facing vibration test conformance reports and justified DUT acceptance criteria across a multi-unit qualification campaign.
- Established a boom stiffness test procedure. Designed and built the test stand, captured deflections via PolyWorks measurement, and correlated the boom FEM to test data within 3%, presenting results to the team and publishing the test procedure.

CU Hyperloop Propulsion Team Member – Mechanical/Aerospace Engineering

Aug 2024 - May 2025

- Designed a high-force soil-gripping system using Kevlar-reinforced inflatable tires rated to 60 kN (FOS 2), enabling controlled underground propulsion.
- Modeled and validated tire performance in CAD/MATLAB, iterating the design using test data and rapid prototyping.
- Contributed to a school-record 0.3 m dig depth and winning the innovation award.

PROJECTS

CU Boulder, Discovery Learning Apprenticeship (DLA) – Aerospace Engineer

Aug 2025 - May 2026

- Modeled cislunar plasma environments in SPIS to develop a concept of operations achieving 70 % higher spacecraft-target charging equilibrium, supporting lower-power touchless debris removal.
- Engineered a charged-lens mounting system in a vacuum chamber, shaping ion-gun beam output to within 1mm of target size.

TECHNICAL SKILLS

Design & Analysis: FEMAP, NASTRAN, FEA, SolidWorks, Fusion 360, Bolted-Joint Analysis, Random Vibration analysis

Fabrication & Testing: Machining (Mill, Lathe, Drill Press, Chop Saw), 3D Printing, Soldering, Test Conformance Reports, Test Procedure Documentation, Shock Testing, Vibration Testing, PolyWorks

Software & Tools: MATLAB, Octave, VibrationVIEW, Git/GitHub, PDM, LaTeX, Excel, Word, Slack, Microsoft Teams